JavaScript Day 3 — Arrays & Objects

Arrays

Store multiple values; indexing starts at 0.

```
let fruits = ["Apple", "Banana", "Mango"];
console.log(fruits[2]); // Mango
fruits[1] = "Grapes";
```

push(), pop(), shift(), unshift()

push adds at end; pop removes from end; unshift adds at beginning; shift removes from beginning.

```
fruits.push("Orange");
fruits.pop();
fruits.unshift("Orange");
fruits.shift();
```

map()

Changes/transforms every item and returns a new array.

```
let numbers = [1, 2, 3];
let doubled = numbers.map(n => n * 2);
// [2, 4, 6]
```

filter()

Selects all items matching a condition.

```
let even = numbers.filter(n => n % 2 === 0);
```

find()

Returns the first item matching a condition.

```
let result = numbers.find(n => n > 1);
// 2
```

reduce()

Combines array values into one final result.

```
let total = numbers.reduce((sum, n) => sum + n, 0);
// 6
```

Objects

Store information as key-value pairs.

```
let student = {
name: "Rahul",
age: 20,
grade: "A"
};
console.log(student.name);
console.log(student["grade"]);
student.age = 21;
```

Object Destructuring

Extract properties into variables.

```
let { name, age, grade } = student;
console.log(name);
```

Spread (...)

Spreads/copies array or object contents.

```
let fruits2 = [...fruits, "Mango"];
let person2 = { ...student, city: "Bangalore" };
```

Rest (...)

Collects multiple arguments into an array.

```
function showEmployees(...employees) {
console.log(employees);
}
showEmployees("A", "B", "C");
```

Quick Memory

map = change every item | filter = select multiple | find = first match | reduce = one result. push/pop = end; unshift/shift = beginning. Spread = spread out; Rest = collect in.

Employee Data Management

Day 3 practical: arrays + objects + methods.

```
let employees = [
{name:"Namratha", age:22, department:"Intern", salary:20000},
{name:"Rohit", age:25, department:"HR", salary:25000},
{name:"Ramesh", age:30, department:"IT", salary:30000},
{name:"Suresh", age:28, department:"Finance", salary:28000}
];

employees.push({name:"Ananya", age:24,
department:"Marketing", salary:22000});

let itEmployees = employees.filter(e => e.department === "IT");
let employeeNames = employees.map(e => e.name);
let highSalaryEmployee = employees.find(e => e.salary > 25000);
let highestSalary = employees.reduce(
(max,e) => e.salary > max ? e.salary : max, 0
);
let {name, department} = employees[0];
let updatedEmployee = {...employees[4], department:"Sales"};

function showEmployees(...employees) {
console.log(employees);
}
```